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Viola seed pod architecture shapes sequential, force-augmented pinching

Cheongsan Kim^{1†}, Jihyun Won^{2,3†}, Donghyeon Kim^{2,3†}, Sohyun Jung^{4*}, Ho-Young Kim^{1*}, Youbong Hyun^{2,3*}

Many plants explosively launch seeds, but these natural catapults often display inefficient, unpredictable energy transfer in seed ejection. Violets (*Viola* spp.) address this problem by ejecting seeds successively with consistent propulsive force from a single pod, a strategy that requires sophisticated energy release. In this work, we show that *Viola* achieves this feat with a simple and compact structure that generates adaptive force augmentation through sequential pinching. Our biological and mathematical analyses indicate that the pod valve's morphogeometry optimizes pinching with sufficient strength for seed ejection with limited material cost and creates a shifting force-amplifying hotspot, which allows consecutive seed ejections. We use this design principle to create autonomous zipping actuators for a range of applications, including biomedical soft machines.

Violets (*Viola* spp.) are successful perennial herbs native to most temperate and subtropical zones of the Northern Hemisphere (1). These plants, like many species, disperse their seeds autonomously through ballistic seed ejection (2). Yet, unlike typical ballistic dispersers that rely on pod explosion to release all seeds at once (3–6), the seed pods of *Viola* eject seeds one after another at substantial time intervals—averaging 5.6 min between ejections in *Viola philippica*. This unusual mode of seed dispersal is driven by the gradual pinching deformation of the pod valve (movie S1), whose presumed function lies in delivering a reliable propulsive force to each seed. As such, the mechanism regularizes energy consumption in seed ejection, thereby avoiding the inefficient and unpredictable payload acceleration common in explosive mechanisms (7–9). *Viola* plants develop these seed pods continuously year-round—averaging 1.6 pods per day in *V. philippica* under a 16-hour light–8-hour dark cycle at 22°C (movie S2)—whereas many other plants restrict it to specific periods. This sustained and effective seed dispersal contributes to the exceptional invasiveness of *Viola* species.

The seed pod of *Viola* consists of three valves that split open at the onset of dispersal (Fig. 1A). Pinching deformation along each valve pushes out seeds that are firmly anchored to the valve's placenta through a connective tissue called the funiculus (Fig. 1B). The compressive force accumulates tensile stress in the funiculus; once this stress surpasses a threshold value, the funiculus fractures, launching the seed at speeds of up to 10 m/s in 1 ms and propelling it several meters away, often more than 20 times the parent's body size (Fig. 1C and movie S3). Similar pinching-based seed ejection has evolved independently in distantly related taxa. However, these seed pods typically release only a single seed, as seen in many natural power-amplification processes (10–12). By contrast, the *Viola* pod executes multiple acceleration cycles during its shape changes until all seeds—~45 to 60 per pod

in *V. philippica*—are discharged. This mechanism requires an intricate, repeated energy release. In this work, we investigated how *Viola* meets these biomechanical challenges using a simple and compact pod structure.

Sequential pinch-coupled seed dispersal

The pinching motion of the *Viola* valve is hygroscopic, triggered by its hydration status. Seed pods of *V. philippica* lose ~20% of their water content in the first 2 hours of dispersal—from 49.1 ± 3.8% to 26.4 ± 5.9% (fig. S1)—which powers the valve pinching (movie S4). Hygroscopic actuation occurs in many natural and bioinspired systems (13–22); however, in sharp contrast to such conventional materials exhibiting global shape changes, the *Viola* valve displays spatiotemporally evolving deformation (Fig. 1D) (23). The valve closes first in distal regions (Fig. 1, D and E), and the pinching progresses toward the proximal end along the valve's long axis. This sequential valve pinching appears conserved among ballistic *Viola* species because similar patterns were observed in other ballistic dispersers, *V. lactiflora* and *V. keiskei* (fig. S2).

Motivated by this distinctive deformation mode, we investigated the mechanical performance of *Viola* valves. Analyzing seed dispersal kinetics and mechanical properties of valve structures showed that ~40% of the potential energy stored in a *V. philippica* seed pod is imparted to the total kinetic energy of expelled seeds (Fig. 1G and supplementary text, section 1). Such a high energy conversion ratio correlates well with the consistent force application for every seed propulsion. In comparison, single-burst explosion mechanisms exhibit markedly lower energy conversion ratios, often less than 1% (Fig. 1G). Indeed, *Viola* achieves this feat with a relatively lightweight pod, whereas explosive dispersers rely on structures that are substantially heavier than the total seed mass, which highlights a material-efficient design of *Viola* pod valves. A similar high energy conversion ratio was also detected in other pinching-based dispersal systems, such as *Hamamelis*, *Fortunearia*, and *Loropetalum* (Fig. 1G). However, these systems disperse only a single seed per pod to concentrate force delivery. Thus, *Viola* reconciles two often competing design constraints in ballistic seed dispersal: energy-regularized and multiseed release (7). These findings prompted us to investigate how the *Viola* valve's architecture enables this functionality, potentially bridged through its distinct sequential pinching action.

Optimized design for pinching

To implement seed propulsion, the force exerted on the funiculus should be greater than its fracture threshold. Thus, we began by studying the pinching force generated by the two-dimensional (2D) deformation of *Viola* valves. Transverse sections (1-mm thick) of *V. philippica* valves (Fig. 2B and fig. S5) revealed two distinct compartments: the arms that contact the seeds in valve closing, thereby transmitting force to sever the attached funiculus, and the semicircular body that drives the pinching motion. Time-lapse imaging showed that during deformation, the body's inner surface contracts by ~40% in length, whereas its outer surface remains largely unchanged (fig. S5 and movie S5). The length change was negligible in the arm sectors (fig. S5). Thus, the differential contraction in the body compartment generates a bending moment that draws the arms together.

We next used tissue microsection (7-μm thick) to show that the body sector consists of two tissue types: the semicircular “responsive tissue” composed of elongated, radially arranged cells (Fig. 2C, r) and the outer “shell” comprising a few layers of smaller, round-shaped cells (Fig. 2C, s). On drying, the isolated responsive tissue underwent substantial shrinkage, whereas the body shell alone deformed minimally (fig. S6). Environmental scanning electron microscopy confirmed that even in the absence of cytoplasmic contents, cell walls in the responsive tissue contract in both length and width with decreasing humidity, identifying this cellular compartment as a primary actuating element (Fig. 2D and supplementary text, section 2). Thus, analogous to many hygroscopic bilayer systems (13, 14), the *Viola* valve body functions as

¹Department of Mechanical Engineering, Seoul National University, Seoul, Korea. ²School of Biological Sciences, Seoul National University, Seoul, Korea. ³Research Center for Plant Plasticity, Seoul National University, Seoul, Korea. ⁴Department of Robotics and Mechatronics, Daegu Gyeongbuk Institute of Science and Technology, Daegu, Korea. *Corresponding author. Email: sohyunjung@dgist.ac.kr (S.J.); hyk@snu.ac.kr (H.-Y.K.); youbong.hyun@snu.ac.kr (Y.H.) †These authors contributed equally to this work.

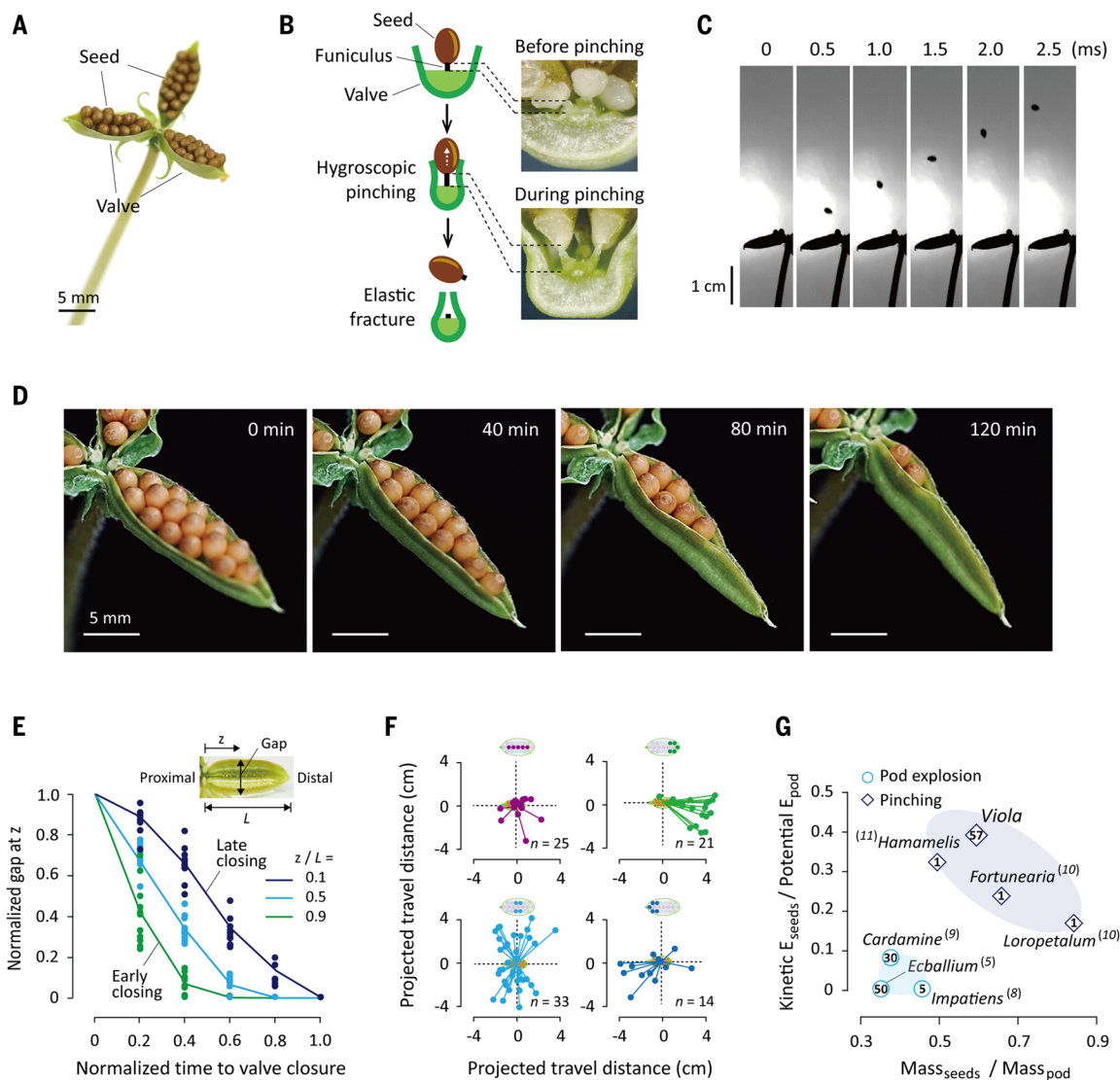


Fig. 1. Sequential pinch-coupled seed dispersal in *Viola*. (A) A photograph of the *Viola* seed pod at the onset of seed dispersal. (B) Schematic illustration of the physical process of seed ejection, with photographs showing elastic strain in the funiculus. (C) High-speed photographs showing a time-lapse sequence of a single-seed ejection from the *V. philippica* valve. (D) Time-lapse images of sequential pinching of the *V. philippica* valve. (E) Dynamics of sequential pinching. Normalized gap distances, defined as the ratio of the gap at a given time point to the initial gap, at different valve positions are shown. Individual data points represent measurements from 10 *V. philippica* valves, each obtained from an independent plant. Solid lines connect the mean normalized gap distances over time. (F) Direction-controlled seed dispersal in *V. philippica*. Individual data points represent the projected seed positions 1/120 s after launch, captured using a camera positioned vertically above the valves. Color-coded seed groups and the number of seeds (n) captured for each seed group are indicated. Data were collected from 10 independent valves. (G) Comparison of mechanical characteristics between seed dispersal strategies. The number inside each data point indicates the average number of seeds per pod. References for data collected from each species are indicated in parentheses.

a composite actuator: The responsive tissue serves as a deforming layer, whereas the body shell acts as a resistant layer, together driving the coordinated pinching in a 2D picture.

Valve pinching varies between *Viola* species depending on their seed dispersal strategies (Fig. 2A) (1, 2, 24). Two contrasting valve behaviors are observed in our species collections (Fig. 2B and movie S6), derived from surveys on natural fields and seed repositories. All examined ballistic *Viola* species, including *V. philippica*, displayed effective valve pinching. Yet, such deformation was greatly attenuated in those species—including *V. odorata*—that merely drop their seeds on the ground to be picked up by seed-transporting ants. Both the semicircular and hygroscopic responsive tissues are present in these valves (Fig. 2C and figs. S6 and S8), despite their discrepancies in pinching capacities.

To elucidate the mechanical basis differentiating these behaviors, we analyzed the deformation of a bilayer structure, composed of semi-circular active inner tissue, with radius R_r and Young's modulus E_r , and a resistant outer shell, with thickness t_s and Young's modulus E_s , having unit depth in the out-of-plane direction (fig. S9). Our model indicates that the hygroscopic contraction of the active layer generates a bending moment, which produces a pinching displacement δ_h in balance with the elastic resistance of both the active inner tissue and the outer shell (fig. S9B and supplementary text, section 3.1). Additionally, owing to the geometrical compatibility of the semicircular bilayer, an additional bending moment can be created through an intrinsic coupling between horizontal and vertical displacements, which drives a geometrically induced pinching displacement δ_g under

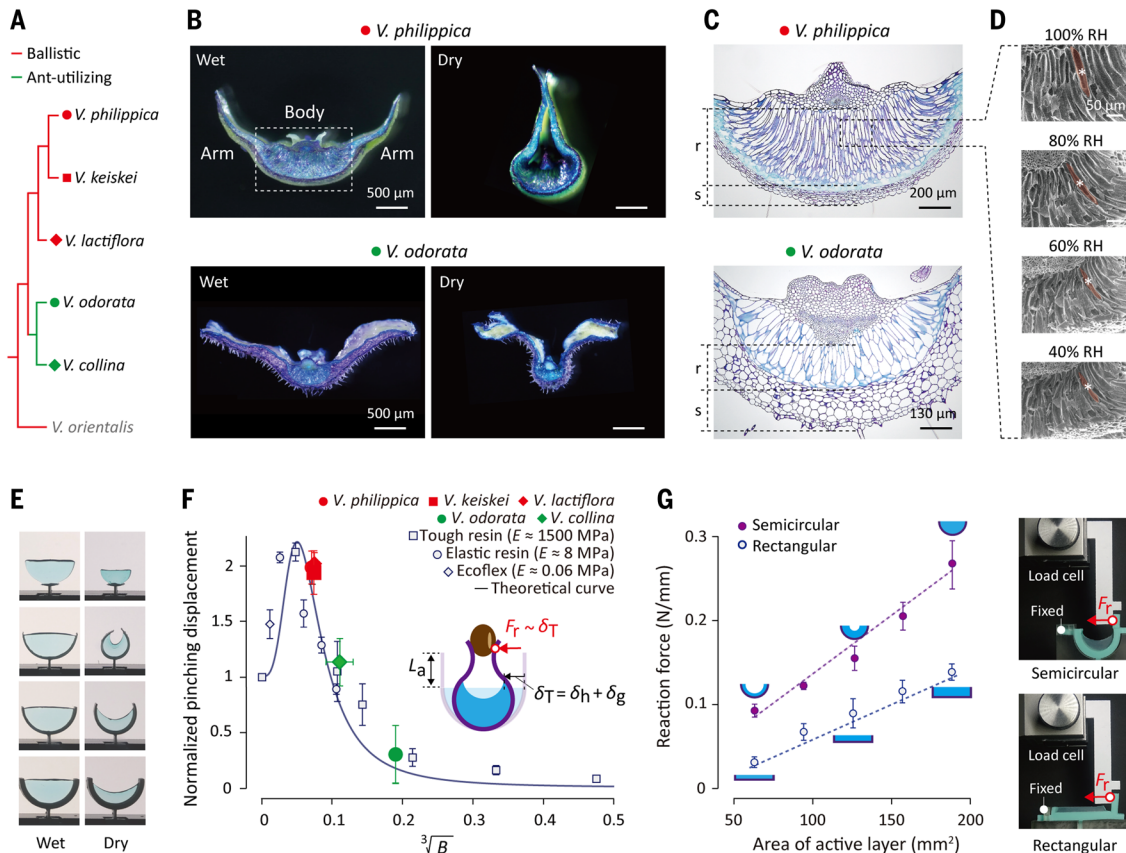


Fig. 2. Optimized semicircular bilayer for strong pinching. (A) Evolutionary divergence of seed dispersal mechanisms in *Viola*, illustrated on a schematic phylogenetic tree adopted from Beattie *et al.* (2) and Yoo *et al.* (24). (B) Variation in the magnitude of pinching between valves associated with differing dispersal mechanisms. Transverse valve sections were stained with toluidine blue O (TBO) to enhance visualization. (C) Histological comparison of the body sector between valves exhibiting varied pinching capacities. Microsections of the valve body were stained with TBO. r, responsive tissue; s, body shell. (D) Humidity-dependent cell wall deformation in responsive tissue. Responsive tissue dissected from the *V. philippica* valve was examined using an environmental scanning electron microscope. Asterisks denote the same cells that undergo shrinkage with decreasing humidity. RH, relative humidity. (E) Experimental validation of pinching capacity. Artificial *Viola* valve bodies with varying thickness of the body shell (black, resistant layer) were fabricated using equivalent responsive tissue (blue, active layer). (F) Optimized architecture for pinching. (G) Semicircular configuration achieving a high force-to-volume ratio. Pinching forces were compared between semicircular and rectangular bilayers fabricated with equivalent material input. The experimental setup for pinching force measurement is shown on the right.

the elastic resistance of the valve body (fig. S9C and supplementary text, section 3.2). The total pinching displacement is given by $\delta_T = \delta_h + \delta_g$ (fig. S9D). Normalizing this value by the characteristic hygroscopic deformation of the responsive tissue $R_t \alpha \Delta \phi$, where α and $\Delta \phi$ denote the hygroscopic expansion coefficient and decrease in water content of the responsive tissue, respectively, leads to the dimensionless pinching displacement

$$\frac{\delta_T}{R_t \alpha \Delta \phi} = \beta_1 \frac{1}{1 + kB} + \beta_2 \frac{kB}{(1 + kB)^2}$$

where β_1 and β_2 are the parameters to be determined experimentally, k is a proportionality constant reflecting the geometric complexities of each layer, and $B = E_s t_s^3 / E_t R_t^3$ corresponds to the bending stiffness ratio of the resistant body shell to the active responsive tissue.

To validate our model, we fabricated artificial *Viola* valves from a hydrogel, 3D-printed resins, and polymers, systematically varying layer thickness and Young's modulus (table S3). These artificial structures demonstrated pinching that is highly sensitive to the thickness ratio between the two layers, t_s/R_t , which greatly contributes to the bilayers' bending stiffness ratio B (Fig. 2E). A theoretical curve (Fig. 2F), with the parameters obtained empirically, exhibited a single peak in the

normalized pinching displacement as a function of $\sqrt[3]{B}$. The resulting apex represents an optimal condition under which the bending deformation is maximized.

As this pinching displacement brings the arm into contact with the seed, the force exerted on the seed per unit depth F_T can be expressed as $F_T \sim E \delta_T t_a^3 / L_a^3$, where t_a and L_a denote the thickness and length of the arm, respectively (Fig. 2F, inset, and supplementary text, section 3.3). Thus, the larger that δ_T is, the stronger that F_T becomes, which enables effective seed propulsion.

On the basis of this insight, we substituted the morphomechanical parameters experimentally determined from the valves of multiple *Viola* species (table S2) into the theoretical model and revealed two distinct clusters. The valves from the ballistic species (Fig. 2F, red markers) were located near the peak, indicating a stiffness ratio that facilitates near-optimal displacement. By contrast, the valves from the species that solely rely on the ant-assisted mechanism (Fig. 2F, green markers) deviated significantly from the optimal regime because of their comparatively thicker outer shells (Fig. 2C and fig. S8), which degrade their bending deformation. Thus, the ballistic *Viola* valves have retained an optimal architecture for pinching over evolutionary time.

The semicircular *Viola* valve body appears during the volumetric growth phase in pod development (fig. S10). Because this configuration

is not common in seed pods of the other *Violaceae* species (25) and man-made bilayers (13), we wondered whether such a developmental cost offers mechanical values in seed dispersal. In conventional rectangular bilayers, the vertical strain (ϵ_v) in the active layer, being perpendicular to the neutral axis of the resistant layer, does not contribute to bending (fig. S11) (26). By contrast, semicircular bilayers harness both radial and circumferential components of the hygroscopic strain owing to the distortion of the neutral axis of the resistant layer (fig S11), thereby resulting in amplified bending moments.

In experimental tests where both actuators shared identical thickness, perimeter, and stiffness of the resistant layer, the semicircular configuration consistently produced stronger forces than the rectangular bilayer at the onset of pinching deformation (Fig. 2G and fig. S11C). As the input of the active layer increased, both designs exhibited

greater force output. However, the force gain was more profound in the semicircular geometry, resulting in a widening performance gap. Thus, the *Viola* valve's semicircular geometry requires less material input to deliver comparable force output compared with conventional rectangular designs. Together, the 2D architectures of *Viola* valves offer a design principle for an effective pinching force generation in bilayer composites.

Sequential pinching and force augmentation

We next turn to the mechanics of sequential pinching, which requires us to consider the 3D *Viola* valve structure. The *V. philippica* valve resembles a sailboat with distinct boundary conditions between the proximal and distal ends (Fig. 3A). The distal end, akin to the sailboat's bow, is free and convex on its underside. By comparison, the proximal

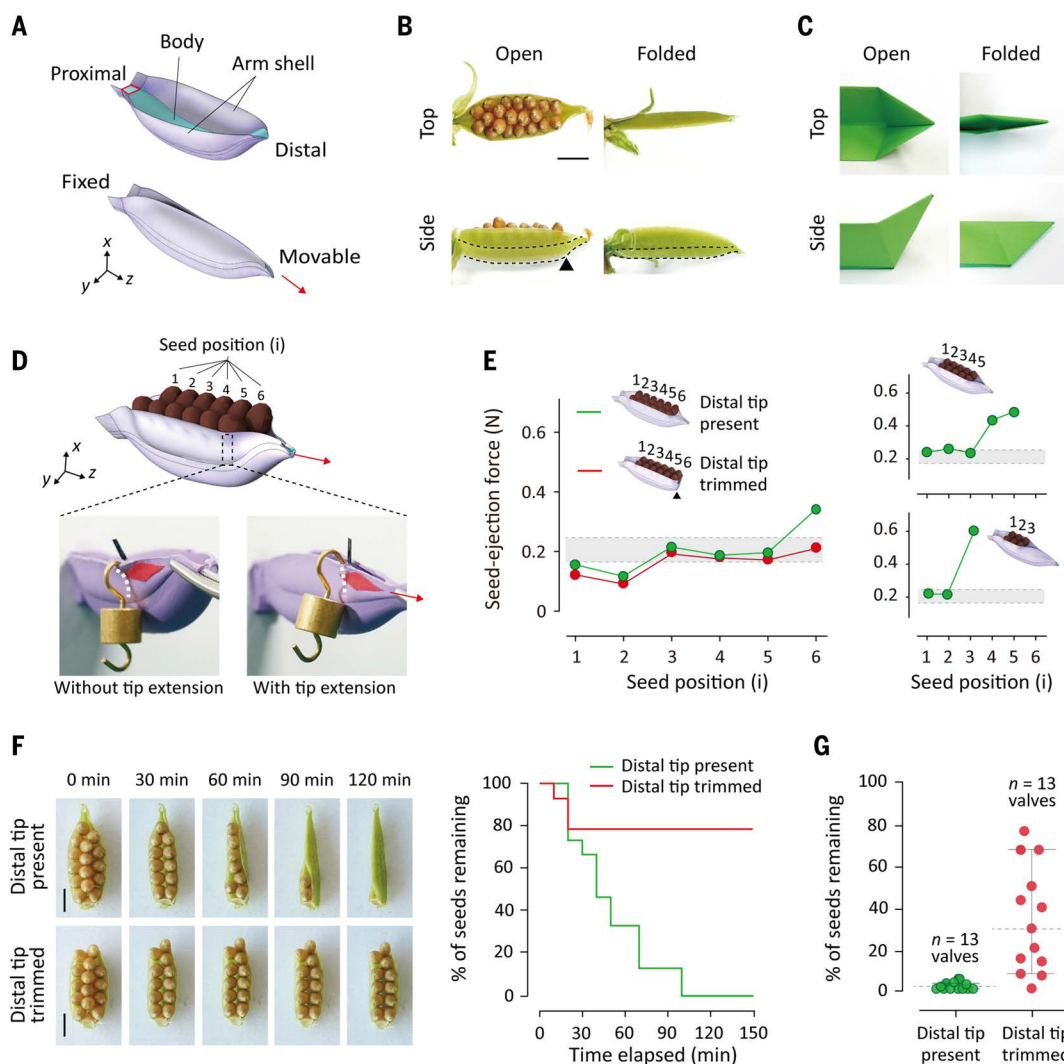


Fig. 3. Force augmentation through sequential pinching. (A) 3D reconstruction of *Viola* valve motion. The red arrow indicates the outward protrusion of the distal tip. (B) Hierarchical deformation of *V. philippica* valve structure. Dashed lines highlight the straightening of the body segment during valve pinching. (C) Origami model replicating the tip protrusion induced by pinching. (D) Tip extension-coupled local arm shell stiffening. White dashed lines visualize the degree of arm shell deflection under applied weight, with and without tip extension. Red arrows denote the distal tip extension. Shape changes of the red rhombus illustrate arm shell stretching in the distal region. (E) Force augmentation mediated by arm shell's local stiffening. (Left) Simulated seed-ejection forces acting along the funiculi attached to the seed at each valve position (i). (Right) Force distribution during sequential progression of valve pinching. Gray boxes indicate the range (maximum to minimum) of fracture threshold measured from the funiculi ($n = 10$) of *V. philippica*. (F) Seed dispersal capacity of *V. philippica* valves under different trimming conditions. (Left) Time-lapse images of seed dispersal. Scale bar, 5 mm. (Right) Seed retention dynamics in *V. philippica* valves subjected to different trimming conditions. Remaining seeds were counted 1 day after trimming. Data were obtained from independently collected valves ($n = 13$ for each condition). The number of seeds per valve ranged from 15 to 25.

end, or the stern, is flatter and fixed to the stem. As the arm shells are pulled toward the central axis with dehydration of the valve body, this curved structure tends to straighten because of the material's incompressibility (Fig. 3B), which in turn generates a force acting along the valve's long axis (z axis).

The structure's ability to deform under this longitudinal force affects the energetic cost of valve closure. In the distal valve, the perimeter of the convex underside (L_d) exceeds its longitudinal projection, providing surplus length that allows the valve segment to straighten in the z axis without substantial stress buildup (fig. S12). The resulting tip motion can be simply replicated using an origami model (Fig. 3C), which demonstrates that this tip motion allows folding about the z axis (movie S7). By contrast, for the proximal valve with a flat underside, insufficient perimeter hinders straightening (fig. S12). Thus, the arm shell closure—when viewed from the top—can be interpreted as the elongation of a curved cantilever beam fixed at the proximal end and free in the distal region (fig. S12).

Consequently, closing near the fixed proximal end (Fig. 3A, red square) demands substantial energy because of the strong mechanical constraints. By contrast, curvature changes at the free distal end require far less energy, causing arm shell closure to take place at the distal end first and then progress toward the proximal end, resulting in the distinct sequential pinching motion of the *Viola* valve (fig. S12). Cellular composition and hydration status did not differ significantly between the distal and proximal valve segments (fig. S12), which indicates that the spatiotemporally evolving valve motion is governed primarily by the geometry.

This anisotropy in boundary conditions biases force generation toward the tip (supplementary text, section 4). As the valve closes, the distal end is primarily drawn into extension, engaging in-plane (membrane) stresses in the distal arm shell (Fig. 3D and fig. S13). Unlike bending stresses, which scale only with the relatively weak bending stiffness Et_a^3 , membrane stresses scale with Et_a , which makes them orders of magnitude stronger in thin-walled structures, such as *Viola* arm shells ($t_a \approx 10^{-4}$ m) (27). This geometric coupling effectively stiffens the arm shell in the distal region and enables such a soft membranous structure to exert a force strong enough for seed ejection. At the same

time, the tapering closure (fig. S14) concentrates contact near the distal apex, further enhancing the local force. Modeling and simulations capture this geometry-driven hotspot (Fig. 3E, left, and fig. S15), where the seed-ejection force acting along the funiculus surpasses its fracture threshold. Consequently, seeds at the distal end are expelled first, and the ejection front progresses proximally as the force-amplifying hotspot shifts (Fig. 3E, right).

To explore effects arising from this spatiotemporally uneven deformation, we compared seed dispersal in *V. philippica* valves subjected to two trimming strategies: one with the distal end trimmed and the other with the distal end retained. With the distal tip present, valves effectively dispersed nearly all seeds through sequential pinching (Fig. 3F and movie S8). By contrast, valves with the distal tip trimmed were markedly less effective in seed ejection (Fig. 3F), often leaving many seeds undispersed (Fig. 3G). Thus, the geometric feature of the distal tip is functionally significant. In line with this, our simulations demonstrate that trimming the distal tip or constraining its extension diminishes force amplification (fig. S15). This, in turn, leads to a more even distribution of force applied to the seeds, thereby keeping it below the fracture threshold of the funiculus (Fig. 3E and fig. S15). The geometry-mediated force augmentation therefore overcomes the energy barrier to execute seed-ejection sequences.

We further considered the ecological correlates detected in explosive seed dispersal mechanisms (7). Although the regularized energy transfer in single-seed shooters enables energy conversion as efficient as that seen in *Viola* (Fig. 1G), they often compromise the spatial diversity of seed distribution owing to the architectural uniformity of seed-ejecting devices. Our model demonstrates that the body straightening reorients the seeds in the *Viola* valve (fig. S16 and supplementary text, section 5), causing them to repel each other, with seeds at both ends adopting a reclined configuration (fig. S16). High-speed video revealed that in *V. philippica*, the seeds positioned furthest from the stem tend to be ejected more horizontally compared with those located proximally (fig. S17) and that the launch directions of the seeds are closely associated with their positions in the valve (Fig. 1F). Together, the 3D motion of the *Viola* valve diversifies seed-ejection trajectories while restricting variation in energy transfer.

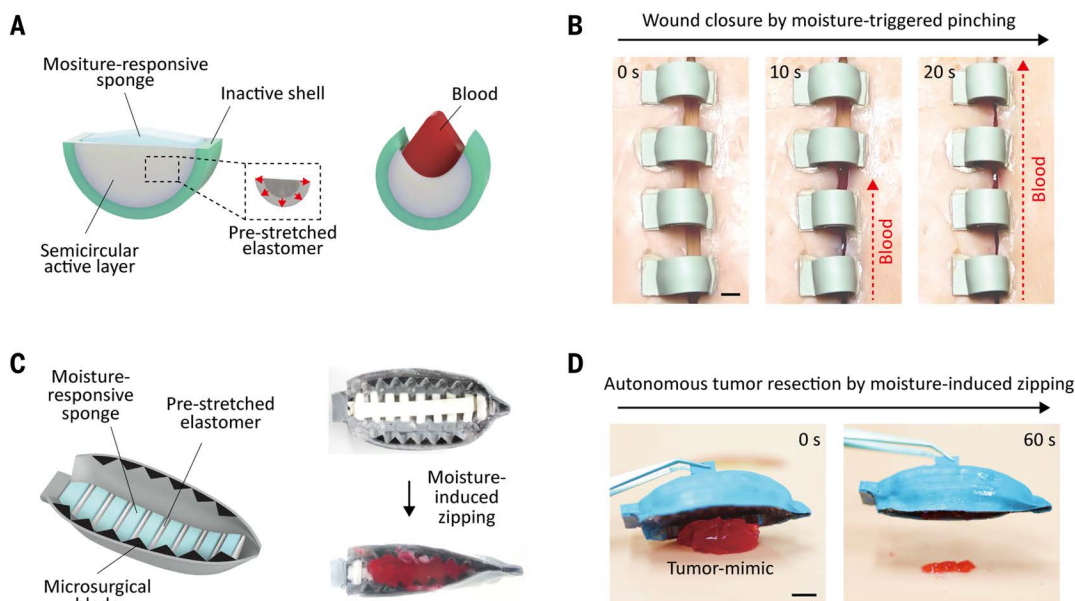


Fig. 4. Geometry-driven, time-tuned zipping mechanism. (A) Moisture-responsive rapid pinching actuators. (B) Moisture-triggered wound closure by pinching actuators. Red dashed lines indicate the extent of artificial blood infiltration in the wound gap of the biomimetic tissue surrogate. (C) Moisture-responsive zipping actuators. (D) Moisture-induced autonomous resection of a tumor mimic enabled by the *Viola*-inspired zipping device.

Geometry-driven, time-tuned zipping mechanism

Inspired by these findings, we developed a pinching mechanism tailored for potential biomedical and related applications. Given that many diffusion-based actuation systems suffer from inherently slow response times (6, 13), we engineered a pinching actuator with acute responsiveness by constructing the device from three components: a semicircular pocket of biaxially prestretched elastomer, a hygroscopic polyvinyl alcohol (PVA) sponge contained in the pocket, and an inactive shell that surrounds the semicircular pocket (Fig. 4A). In the dry state, the high stiffness of the inner sponge counteracts the bending moment induced by the prestretched elastomer pocket, thereby deterring actuation. On hydration, the sponge rapidly absorbs moisture through capillary action, with its modulus decreasing markedly—from ~10 MPa to as low as 5 kPa—thus allowing the prestretched pocket to shrink, which triggers the pinching actuation of the system (Fig. 4A). Unlike its biological counterpart that shrinks when dry, our artificial actuator shrinks when wet—a design choice made to enable rapid actuation across a wider range of applications. This pinching mechanism (Fig. 4B) enabled timely and rapid sealing of wound gaps in a biomimetic tissue surrogate as the sponge in the pinching actuator absorbed artificial blood from the wound site (closure of each actuator occurred within 5 s upon exposure to moisture) (movie S9).

We also developed a zippering device comprising multiple pinching actuators aligned in a soft shell that emulates the 3D architecture of *Viola* valves (Fig. 4C). Unlike other synthetic hygromorphs whose sequential motion is programmed by differential hygro-responsiveness (28), the stepwise pinching in our device was governed solely by geometry. To harness this soft zipping, microsurgical blades were affixed along the upper edges of the arm shells (Fig. 4C). On wetting, the device autonomously resected a tumor mimic made of gelatin with great ease (Fig. 4D). Such a geometry-driven, stress-mitigating excision mechanism could be especially valuable in minimally invasive, robot-assisted, or endoscopic surgeries, where narrow anatomical corridors or proximity to vital structures make conventional scalpel-based resection technically difficult (29, 30).

Conclusions

We investigated how the compact *Viola* valve structure supports high-speed, successive seed ejection—a delicately programmed function enabled by embodied physical intelligence (supplementary text, sections 6 and 7) (31, 32). *Viola* achieves this feat through a simple yet ingenious valve design that yields sufficient force output without the additional biological cost of increasing pod size balancing with other physiological demands (33). The *Viola* pod thus provides an architectural motif for enhancing the mechanical performance of shape-morphing structures with limited intrinsic material properties (34). Guided by this motif, we engineered actuators that recreate the sequential pinching of *Viola* valves and expanded their capabilities to respond to moisture. Taken together, these results demonstrate a virtuous cycle in which mechanics and geometry elucidate biological function, whereas biology inspires physically intelligent architectures that remain difficult for human or computational methods to conceive.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Supplementary Text; Figs. S1 to S19; Tables S1 to S3;
References (36–52); MDAR Reproducibility Checklist; Movies S1 to S9
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Cheongsan Kim, Jihyun Won, Donghyeon Kim, Sohyun Jung, Ho-Young Kim, and Youbong Hyun

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Editor's summary

Many plant species use ballistic seed dispersal, meaning that seeds are released all at once through pod explosion. By contrast, the perennial herb violet (*Viola*) ejects seeds one after another. Kim *et al.* found that violets achieve this through a sequential pinching of the seed pod valve (see the Perspective by Patek). The violet's pod valve geometry optimizes pinching from its farthest to its closest end. Resulting shifting force utilization enables violet valves to overcome the energy barrier for consecutive seed ejection. Using this design principle, the authors engineered a device that shrinks when wet, with potential applications in minimally invasive surgeries. —Unnati Sonawala and Madeleine Seale

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